



GENE STORM

Game Design Document

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1 Game Overview

1.1 Elevator Pitch

Working Title: C.H.A.L. - Customized Hero Arena Looter [TBD]

An auto-battler / team-builder with a third-person top-down camera in which the player acts as a *team architect* and *build alchemist*. Battles play out fully automatically, while the player optimizes their heroes, items, and skill trees like in an ARPG and becomes permanently stronger through meta-research.

Maps consist of increasingly difficult waves, higher difficulty tiers, and better loot — without classic gear grind and without permanent loss of progress after many hours of play.

“A chill auto-battler where you obsess over builds, not button presses.”

Tags

- Genre: Auto-battler, team-builder, ARPG-style progression
- Structure: Wave- and map-based, increasing difficulty tiers
- Perspective: Third-person, top-down, observer role
- Progression: Character levels, items, skill trees, meta-research
- Mode: Primarily singleplayer

1.2 Product Summary

Aspect	Description
Genre	Auto-battler / team-builder with ARPG-style character progression
Core fantasy	Player as team architect and build alchemist who creates ever more powerful hero combinations
Structure	Maps with increasing difficulty, divided into waves
Camera	Third-person / top-down, no direct control of heroes
Controls	Focus on preparation (builds, crafting, research), battles run automatically
Sessions	Alternation between focused planning phases and relaxed spectator phases
Types of progression	Hero levels, items, skill trees, deterministic crafting, meta-research
Platform (planned)	PC (option: mobile port later)
Business model	Buy-to-play, with optional cosmetic-only skins as microtransactions

Table 1: Product overview



1.3 Vision & Design Goals

Design Pillars

- **Relentless progression without micromanagement:** The player experiences a clear, tangible increase in power without having to constantly issue commands during combat.
- **Tactical depth through team and build combinations:** Complexity lies in the choice and synergy of heroes, items, and skill trees, not in APM.
- **Every build should be “playably good”:** There are weaker and stronger combinations, but ideally no build dead-ends that feel like traps.
- **Deterministic / controllable crafting instead of pure gear grind:** Progress is predictable; the player works toward concrete goals.
- **Clear loop of focus and reward:**
 - Focus phase: Crafting, team-building, skill allocation, research.
 - Reward phase: Watching auto-battles, enjoying effects, collecting loot.

Anti-Goals

- **No hard loss of long-term progress:** No systems where the player feels they have “lost everything” after many hours (e.g. permadeath across the entire meta-progression).
- **No action-heavy high-APM gameplay:** Player skill lies in planning and pre-combat decisions, not in reaction time or precise timing during combat.
- **No braindead gear grind:** Purely farming the same content for tiny loot upgrades should be avoided; progress should come from systems like research and deterministic crafting.
- **No pay-to-win mechanics:** Monetization (if present) is limited to cosmetic options.

1.4 Target Audience

Primary Audience

The primary target audience are players with limited time who still want a strong sense of progression and reward:

- **Casual to midcore players** who have little interest in endless grind.
- Players with a **high need for reward feedback** (endorphin hits from visible progress, new builds, clear power spikes).
- Players who enjoy alternating **concentration phases** (planning, optimizing) with **relaxation phases** (watching, cashing in loot).

In short: *“Not much time, but a craving for strong builds and a satisfying power fantasy.”*



Player type	Description	Motivation	Risks
Progression junkie	Plays a few hours per week but expects noticeable jumps in progress.	Fast power spikes, visible upgrades, unlocking new maps/waves.	Frustration if progress is too slow or too RNG-heavy.
Build tinkerer / team architect	Likes to experiment with hero combinations, skill trees, and items.	Finding synergies, discovering “secret” strong combinations.	Overwhelm if systems are too complex or full of build traps.
Relaxed spectator	Uses battles as a “lean back and watch” phase.	Visual feedback, effects, clear understanding of why something wins/loses.	Long downtimes or unclear fights without an obvious reason for victory/defeat.

Table 2: Target audience overview

Player Type Overview

1.5 Product Positioning & Reference Titles

References

- **Backpack Battles / Teamfight Tactics** (game feel)
- **Path of Exile 2** (depth of progression, skill trees, loot fantasy)
- **Research / battle pass systems** (meta-progression without the monetization component)

Differentiation

Compared to these references, the game focuses on:

- **No active player skill during combat:** In contrast to many ARPGs, the emphasis is not on manually executing skills but on preparing and configuring heroes and builds.
- **Semi-deterministic crafting with small RNG layer instead of loot lottery:**
 - The player can deliberately work toward specific items and effects.
 - Crafting targets are deterministic, but rolls within a controlled range keep items exciting without derailing progression
 - Randomness exists but should not be the dominant factor for progression.
- **Every build should be “viable”:**
 - There are stronger and weaker setups, but ideally no completely dead options.
 - The goal is to encourage experimentation instead of punishing it with trap choices.

1.6 Platform & Business Model

Platforms

- **Primary:** PC
- **Option:** Later mobile port if UI/UX and performance make it sensible.



Business Model

- **Buy-to-play:** One-time purchase for the full game.
- **Optional monetization:** Cosmetic-only skins / visual customization for heroes or effects.
- **No-gos:** No pay-to-win mechanics, no loot boxes with gameplay-relevant advantages, no mandatory battle pass.

1.7 Scope & Constraints

Team & Resources

- **Development:**
 - 1 lead developer (game design, programming, playtesting).
 - AI support for coding, testing, and documentation.
- **Art & content:**
 - Asset purchases for 3D models, animations, and textures.
 - Focus on reuse and systemic designs instead of massive one-off content production.

Risks and Constraints

- **Ambitious system depth:** Auto-battler, ARPG progression, deterministic crafting, and meta-research together are very demanding for a solo project.
- **Balancing effort:** The claim that “every build should be reasonably strong” makes balancing and content creation significantly more complex.
- **Art and UX effort:** Third-person top-down with clearly readable auto-battles requires good readability, effects, and UI communication, even when relying on purchased assets.

2 Core Gameplay Loop

2.1 High-Level Core Loop

The central game flow is divided into two clearly separated phases:

- **Active loop (hub / preparation):** The player makes decisions, builds their team, and improves their resources.
- **Passive loop (map / combat):** The player watches auto-battles, collects loot, and fuels the next active loop with it.

Hub Structure

The player starts each session in the **hub**, which contains several station elements. To reduce early complexity, these stations are **introduced in tiers** over the course of the game:

- **Tier 1 – Core loop stations (always available early):**



- **MapDevice:** Select map and difficulty / wave configuration, start the map (= transition into the passive loop).
- **HeroPanel:** Core hero management:
 - * Level up heroes (passive bonuses).
 - * Assign / respec skills.
 - * Change gear.
 - * Team composition (who participates in the map?).
- **Tier 2 – Progression and item depth:**
 - **CraftingBench:** Use loot from previous runs to craft stronger gear.
 - **ResearchLab:** Select an active research project; progress is generated through kill counts (monster-type-sensitive), map completions (difficulty-sensitive), and similar metrics.
- **Tier 3 – Advanced optimization systems:**
 - **ReforgeStation:** Enhance existing gear with new affixes using a meta-currency; intended as a late-early-/midgame optimization tool rather than a starter system.

High-Level Loop

```
[Hub / Active Loop]
-> CraftingBench / ReforgeStation
-> ResearchLab (select/continue research)
-> HeroPanel (level, skills, team build)
-> MapDevice (select map + difficulty)
[Map / Passive Loop]
-> Play waves (auto-battle, loot, XP)
-> Map completed or failed
-> back to hub (Active Loop)
```

2.2 Combat Loop (per Wave)

The combat loop describes the flow within a single wave on a map. Battles are fully automated; player decisions happen before the wave starts.

Before the Wave Starts

- The player **positions their heroes** on fixed slots on the map.
- Positioning is part of the tactical layer (e.g. frontline/backline, AoE positioning, protecting fragile heroes).

Wave Flow

1. **Wave start:** The enemy wave is spawned automatically based on the map configuration.
2. **Auto-battle:**
 - Heroes and enemies act fully autonomously; there is no direct control by the player.
 - Heroes use their abilities, movement, targeting, and AI behavior without input.



3. Loot drops:

- Enemies drop loot.
- The player can actively pick up loot during or after the wave (*reward center trigger*).

4. Experience gain:

- Heroes gain XP for a successfully completed wave.
- If a hero dies during the wave, that hero loses the XP from this wave.
- There is **no auto-heal between waves**; healing only occurs via skills/heroes that can actively heal.

Win / Loss per Wave and Map

▪ Wave won:

- The wave XP for surviving heroes is saved.
- The wave's loot can be collected and used later in the hub.
- The player can:
 - * proceed to the next wave, or
 - * end the run after the wave and return to the hub (progress up to the last successful wave is preserved).

▪ Wave lost (all heroes dead):

- The current wave counts as failed.
- Loot, gold, and XP from this wave are lost.
- The map can be attempted again (optionally after adjustments in the hub).

▪ Map won (all waves defeated):

- The map is marked as *completed*.
- New difficulty tiers for this map are unlocked.
- Completion rewards are granted:
 - * meta-currency,
 - * progress in research / battle-pass-like systems,
 - * possibly special one-time rewards (e.g. unlocking new heroes or maps).

2.3 Map Structure

Wave and Enemy Configuration

- Each map has a **fixed wave structure**:
 - Number and order of waves are predefined.
 - Waves can contain different enemy types, enemy ranks, and tags.
- Within a map, the enemy structure is **semi-fixed**:



- Base compositions are defined via configuration data (enemy ranks, types, tags).
- There is room for variation through modifiers (e.g. meta-progression, map mods).
- Typical structure (configurable):
 - Early waves with standard and normal enemies.
 - Mid waves with increased magic/elite enemies.
 - Late waves with bosses or heavily buffed enemy combinations.

Map Difficulties

- Each map features several **difficulty tiers**.
- Unlocking occurs:
 - by completing the previous difficulty,
 - and/or through progress in research as an additional gating mechanism.
- Higher difficulties offer:
 - tougher enemies (more health, more damage, new enemy types),
 - better loot and stronger progression rewards.

2.4 Research & Meta-Progression in the Loop

Research Activation

- Research is *actively selected* in the **ResearchLab** in the hub.
- Per unit of time/run, one or several research tasks may run in parallel (design detail to be refined later).
- Progress occurs through:
 - kill counts that differentiate by monster types or tags,
 - completed maps and difficulties,
 - optionally further in-game metrics (e.g. resources spent, mechanics successfully used).
- A **HUD element** (e.g. top left) always shows the current progress of the active research.

Research Rewards (Unlocks)

Research systematically unlocks new elements and serves as a guide through the system landscape:

- **New hideout elements:**
 - Stations or features in the hub are unlocked step by step (tutorial-like).
- **New heroes:**
 - Access to additional hero types, archetypes, or specializations.



- **New item tiers:**
 - Higher quality levels and stronger base or affix values.
- **New crafting options:**
 - Additional reforge methods, specialized crafting recipes, more deterministic upgrades.
- **New map tiers:**
 - Linear unlocking of new maps.
 - Unlocking higher difficulties within existing maps.

2.5 Session Structure (Planning vs. Observation)

Typical Planning Phases in the Hub

- **Standard planning (2–5 minutes):**
 - Small crafting decisions.
 - One level-up and skill point allocation.
 - Minor gear adjustments.
 - Back to the map.
- **Planning phase “Changes” (5–10 minutes):**
 - Introducing a new hero into the team.
 - Significant changes to gear and skills.
 - First tests of new synergies.
- **Planning phase “Makeover” (≈ 20 minutes):**
 - Complete team overhaul.
 - Redistribution of skill points.
 - Gear optimization towards a new target build.

Typical Battle Phases on Maps

- **Early game:**
 - ~ 5 minutes.
 - approx. 5 waves.
- **Midgame:**
 - 5–10 minutes.
 - approx. 10–20 waves.
- **Early endgame:**
 - 10–20 minutes.



- 10–20 waves at higher difficulty.
- **Late endgame:**
 - 20+ minutes.
 - hardest maps; an *endless mode* option is planned.
- After **each wave** (successful or failed), the player can exit.
- Progress is saved up to the last successfully completed wave.

Phase	Typical duration	Player activity	Cognitive load
Hub: standard planning	2–5 minutes	small tweaks (crafting, level, gear)	medium
Hub: Changes	5–10 minutes	add new heroes, test builds	high
Hub: Makeover	≈ 20 minutes	complete team overhaul	very high
Map: early game	~ 5 minutes	watching, picking up loot	low
Map: midgame	5–10 minutes	watching, reading early challenges	low–medium
Map: endgame	10–20+ minutes	critical fights, loot decisions	medium

Table 3: Session structure: planning vs. battle phases

2.6 Failure & Recovery

Failure Philosophy

- **Core stance:** Failure should be a *small, manageable pain* that signals:
 - “Okay, I need to update my build.”
- No hard resets on the meta layer:
 - Research, gear, skills, hero levels, and map progress are **permanent**.
 - Defeats primarily affect the current map/wave and the short-term rewards (XP/loot from that wave).

What Is Always Preserved?

- **Research** (research progress and unlocked systems).
- **Gear inventory** (already crafted/looted items).
- **Skills and hero levels** (including unlocked skills/passives).
- **Map progress** (unlocked maps and difficulties).



2.7 Loop End & Long-Term Motivation

Natural Session Endpoints

A session typically ends when one or more of the following points are reached:

- The player has **enough loot** to aim for the next relevant gear upgrade.
- The player has **enough XP** for significant level-ups of individual heroes and thus unlocks new skills/passives.
- New **maps or heroes** are unlocked and form clear “next goals”.
- A major build goal (e.g. a specific archetype or synergy) has been reached or is close.

Long-Term Motivation

In the long run, the player is motivated by:

- Achieving **ever stronger builds** through crafting, reforging, and research.
- **Unlocking new heroes** and testing new team compositions.
- **Beating new maps and higher difficulties** as clear progression markers.
- Experimenting with **build variants** that differ in strength but all remain playable.

3 System Specifications

3.1 System Overview

This section provides a high-level overview of all core game systems and how they connect to the core loop. Each system is defined by its main responsibility, its primary inputs and outputs, and its relation to other systems.

3.2 System Responsibilities

This subsection briefly defines what each system is responsible for and what it explicitly does *not* handle. Detailed mechanics are described in the dedicated sections (e.g. Skill System, Items and Inventory, Crafting System).

Research System

3.2.0.1 Responsibility

The Research System converts long-term play (XP, kills, completed Maps) into unlocks that expand the game: new Hideout elements, Heroes, Item tiers, Crafting options and higher Map difficulties. It controls the pacing at which complexity and power are introduced.

3.2.0.2 Non-Goals

- Does *not* grant temporary combat buffs for a single run or Map.
- Does *not* decide loot quality directly (it only unlocks higher tiers or new sources).



Skill System & Crown System

3.2.0.3 Responsibility

The Skill System defines each Hero's active abilities and passive effects based on gained XP and allocated points. Hero Passives modify base skills (damage, behaviour, synergies) and shape the role of a Hero within a team.

The Crown System provides an additional endgame layer of powerful, rare upgrades that sit on top of the normal Skill/Passive progression and offer long-term goals for highly invested players.

3.2.0.4 Non-Goals

- Does *not* manage gear or item stats (handled by Items & Inventory / Crafting).
- Does *not* spawn or control enemies (handled by Map & Combat System).

Inventory System

3.2.0.5 Responsibility

The Inventory System stores and organises all items and currencies the player owns: gear pieces, crafting materials, gold and rare currencies. It defines capacity rules (if any), sorting/filtering behaviour and how items are transferred between sources (loot, rewards) and sinks (crafting, selling, upgrading).

3.2.0.6 Non-Goals

- Does *not* generate items (handled by Loot System and Crafting System).
- Does *not* define item stats or balance (defined in the Items and Crafting sections).

Crafting System

3.2.0.7 Responsibility

The Crafting System consumes items, recipes and currencies to produce new or upgraded items. It covers three main mechanic families:

- **Recipes Crafting:** Transforming inputs into specific item outcomes.
- **Upgrading:** Improving existing items (tiers, affix strength, implicits).
- **Hardening:** Locking in or protecting certain properties at a cost.

3.2.0.8 Non-Goals

- Does *not* govern drop chances or enemy rewards (Loot System).
- Does *not* track ownership or placement of items (Inventory System).

Map & Combat System

3.2.0.9 Responsibility

The Map & Combat System takes a configured team of Heroes (including their skills and gear) and a chosen Map setup, spawns Waves of enemies and executes auto-battles. It decides:

- when a Wave or Map is won or lost,
- which combat events trigger loot drops,
- how much XP each Hero gains per successful Wave.



3.2.0.10 Non-Goals

- Does *not* decide what exact items drop (item generation is part of the Loot System).
- Does *not* change hero builds or gear (handled in the Hub by Skills/Crafting/Inventory).

Loot System

3.2.0.11 Responsibility

The Loot System transforms combat events (e.g. enemy deaths, Map completion) into concrete rewards: items, gold, rare currencies. It controls rarity distributions, item tier chances and special drops, and feeds both the Inventory and Crafting Systems.

3.2.0.12 Non-Goals

- Does *not* execute combat or determine victory/defeat.
- Does *not* handle long-term unlocks (handled by Research and Meta Progression).

Meta Progression / Battlepass

3.2.0.13 Responsibility

The Meta Progression / Battlepass System tracks long-term objectives across multiple sessions (e.g. completed Maps, Waves, specific challenges) and rewards them along a defined track, primarily with cosmetic rewards and light account-wide bonuses.

3.2.0.14 Non-Goals

- Keine Pay-to-Win-Effekte: keine direkten, dominanten Combat-Vorteile an Echtgeld gekoppelt.
- Steuert keine Kernprogression der Helden (XP, Skills) oder der Items (Tiers, Affixe).

4 Research System

4.1 Overview

The Research System provides long-term, account-wide progression by converting play (XP, kills, completed Maps) into unlocks. It controls when new systems, heroes, item tiers and world variants become available and thus paces both power growth and complexity.

Research is organised into multiple *research trees*. For the MVP, three main trees are planned:

- **Crafting Tree** – unlocks and deepens all item- and crafting-related systems.
- **World Tree** – unlocks Map modifiers and enemy-ecosystem variants.
- **Heroes Tree** – unlocks new Heroes, skill tiers and the endgame Crown system.

Inputs to the Research System:

- XP (global or research-specific, derived from playtime and Map completions),
- Kill counts (monster-type / tag sensitive),
- Completed Maps and Difficulties.



Outputs from the Research System:

- Unlocks of systems and content (Hideout elements, Heroes, item tiers, crafting options, Map tiers),
- Access to higher difficulties and new enemy variants,
- Endgame enhancement layers (e.g. Crown sockets).

4.2 Research Trees (MVP Scope)

For MVP, the Research System is structured into three primary trees: *Crafting*, *World* and *Heroes*. Each tree focuses on a specific axis of progression and unlocks both systems and content along that axis.

4.2.0.1 Crafting Tree

The Crafting Tree defines how quickly and in which order the player gains access to more advanced item manipulation. It starts with basic gear usage and gradually unlocks more powerful and more deterministic crafting tools.

4.2.0.2 World Tree

The World Tree focuses on Maps and enemy ecosystem variants. It controls how dynamic, dangerous and diverse the world becomes.

4.2.0.3 Heroes Tree

The Heroes Tree unlocks new Heroes and deepens their progression layers. It is tightly coupled to the Skill System and the Crown System and must be paced carefully to avoid overwhelming players too early.

4.3 Research Progression and Pacing (MVP-Level)

For MVP, the Research System should:

- Introduce **Crafting basics** and **basic Map empowerment** early, to make loot and Maps meaningful quickly.
- Gate **World variants** (undead/holy) and **endgame crafting (Gear III, Infuse III, Over-Tempering)** behind mid- to late-game play.
- Unlock **Hero Packs** and **Skill tiers** step by step, so that players first get to know their builds, before they Crown-Sockets come in to the game.

5 Inventory System

5.1 Overview

The Inventory System manages all player-owned resources outside of combat. It is split by item type into dedicated sub-inventories, but presented as a single, tabbed UI inside the Hideout.

- **Per item type a dedicated inventory:** each item family (Parts, Remains, Runes, Gear, Modules) has its own capacity and stack behaviour.



- **Single Inventory Menu:** all sub-inventories are accessible via tabs in one consolidated screen for clarity.
- **Hero Gear Slots:** each Hero has 6 equipment slots that are filled from the Gear inventory:
 - Head, Chest, Gloves, Leg, Boots, Amulet.
- **Hideout-only Access:** inventory management (equipping, moving, crafting usage) is only available in the Hideout, not during Maps.

Inventory Categories:

- *Items* (Gear, Modules, Runes): Individual, building-related objects with identity and decision-making quality.
- *Materials* (Parts, Remains): Stackable resources that primarily serve as input for crafting and upgrading.

The Inventory System itself does not generate items or currencies and does not determine drop chances. It acts as the persistent container and organiser for all loot generated by the Loot System and all results created by the Crafting System.

5.2 Resource Types

The game distinguishes several item families and currencies. Each family has its own stack rules and usage pattern.

Item Types

Type	Description	Stack Limit	Primary Source	Primary Use
Parts	Leftover body parts and fragments from monsters; main generic crafting material.	100	Enemy drops (all Maps)	Baseline crafting inputs for gear and other items.
Remains	Monster drops representing meta-materials in various rarities (e.g. ash, glitter dust, boss cores).	999	Enemy drops (rarity scaling with monster/map difficulty)	Advanced and meta-level crafting operations; higher-tier recipes and reforging.
Runes	Craftable endgame items.	10	Crafting results, rare drops	Socketed or slotted into systems (e.g. endgame build customisation).
Gear	Equipable items for Heroes (weapons/armour/charms, abstracted into slots).	1 (no stacking)	Crafting results, rare drops, rewards	Direct Hero power via equipment: stats, affixes, unique effects.
Modules	Craftable/findable items for configuring and augmenting skills.	10	Crafting, targeted drops, rewards	Modify skills: behaviour, triggers, numbers, interactions.



Currencies

▪ Gold

- Main currency.
- Almost all Hideout activities (crafting, reforging, respects, Map-device usage, etc.) cost Gold.
- Acts as the primary brake against endless crafting loops.

▪ Crystals

- Secondary, rarer currency.
- Placeholder for high-end or structural unlocks (e.g. endgame systems, Hero unlocks).

5.3 Inventory Structure & Rules

Global Inventory and Sub-Inventories

- **Single player inventory:** currently, each player has exactly one main inventory; there are no separate stash tabs or alternate character inventories.
- **Per-type sub-inventories:** within this main inventory, each item type has its own sub-inventory with:
 - its own capacity model (stack limits, number of stacks),
 - type-specific UI representation (tab, iconography),
 - type-specific sorting/filtering options (future).
- **Hero Equipment:**
 - Heroes do not have separate bags; they only have 6 equipment slots (Head, Chest, Arm, Leg, Boots, Charm).
 - Equipping and unequipping gear always transfers items between Hero slots and the global Gear inventory.

Access and Restrictions

- **Hideout-only:**
 - Inventory management is only available in the Hideout (Hub).
 - During Maps, the player cannot:
 - * move items,
 - * change gear,
 - * initiate crafting/recycling actions.
- **During Maps:**
 - Items and currencies are collected, but not manipulated.
 - Gear decisions and build changes are postponed until return to the Hideout.



5.4 Loot Pickup & Interaction

Drops

- **Enemies drop:**
 - Parts,
 - Remains,
 - Currencies (primarily Gold).
- **Drop presentation:**
 - Loot appears as neutral bags on the ground.
 - Bags are colour-coded by rarity.

Pickup Behaviour

- **Manual Pickup:**
 - The player clicks on loot bags on the screen.
 - Each pickup triggers visual and audio feedback, plus a small text log at the side summarising what was collected.
- **Optional „Pick All“:**
 - After each Wave, an optional „Pick All“ function can collect all remaining loot at once.
- **Currencies:**
 - Currency drops from enemies are collected automatically on pickup (no separate item objects).

Loot Usefulness & Filtering

- **No loot filter:**
 - There is no explicit loot filter planned.
 - Design assumption: *every drop is useful* as input into crafting or progression, even if indirectly.

5.5 Economic Role of the Inventory

The Inventory System itself is intentionally designed as a *comfort container*, not as a primary economic pressure tool:

- **No hard space pressure:**
 - The main friction in the economy is not inventory space, but the availability of currencies and materials (especially Gold and high-tier Remains).
 - Item stack limits are present but relatively generous, mainly to maintain readability and avoid infinite hoarding without cost.
- **Economic pressure via costs, not slots:**
 - Gold acts as the main brake on crafting and Hideout actions.
 - Rare currencies (Crystals, late-game Remains) gate high-impact upgrades and endgame systems.



6 Map System

6.1 Overview

The Map System provides the combat stage for all auto-battles. Each Map is an arena-style combat area with a fixed Wave structure and one or more Difficulty levels. The MapDevice in the Hideout acts as the entry point: the player selects a Map, a Difficulty (if unlocked) and optionally empowerment modifiers before starting a run.

Maps are tightly integrated with:

- the **Hero & Skill System** (which team and build enters the Map),
- the **Loot System** (which rewards are generated from enemies and completion),
- the **Research System** (which Maps and Difficulties are available, and which enemy variants exist),
- the **Inventory System** (where all drops are ultimately stored).

The Map System itself does not generate items or currencies directly; it defines when and how enemies spawn and which combat events feed into the Loot System.

6.2 Map Types and Layout

Arena Structure

- Maps are **arena-like**: one main combat area per Map.
- There is no free player movement during combat; heroes and enemies move and act autonomously within the arena.
- Within this arena, different **layouts and obstacles** exist per Map:
 - terrain features,
 - obstacles,
 - lanes and choke points.
- Layout influences:
 - hero positioning decisions,
 - pathing and clustering of enemies,
 - effectiveness of area-based skills and auras.

Themes and Map Variety

- For MVP, the game targets **3 high-level themes** (e.g. different fantasy / dungeon styles).
- Per theme, there are **3–5 Maps** with distinct terrain/obstacle layouts:
 - visual variation (biomes, architecture),
 - gameplay variation (different chokepoints, open areas, spawn positions).
- In total, MVP aims for **around 10 Maps** with varied Wave setups and enemy mixes.



6.3 Map Selection (MapDevice)

Selection Flow

From the MapDevice in the Hideout, the player selects:

- **Map:** any Map that has been unlocked via Research or previous completions.
- **Difficulty:** a Difficulty level for that Map (if unlocked).
- **Wave Range (optional):** within a Map, specific Wave ranges or starting points may be selected if previously reached.

Maps are presented as a **fixed list of unlocked Maps**, not as a rotating offer. Map and Difficulty are selected separately: a Map has a base identity (theme, layout, enemy families), while its Difficulties add scaling and modifications.

6.4 Wave System

Wave Structure

- Each Map has a **fixed Wave structure**:
 - total number of Waves,
 - base composition per Wave (enemy types, ranks, tags),
 - temporal pacing (early/mid/late Waves).
- Wave blueprints are **finely configurable**:
 - relative amount of basic vs. Magic vs. Elite enemies,
 - which Waves can spawn Bosses,
 - where higher-density or special packs appear.
- Design guideline for early MVP Maps:
 - early Waves: mostly basic enemies, low pressure,
 - mid Waves: increasing number of Elite enemies and more complex packs,
 - late Waves: Bosses and high-pressure encounters.

Bosses and Champions

- **Bosses:**
 - are boosted versions of regular units with significantly higher stats,
 - have more skills and more complex behaviour,
 - typically appear in later Waves of a Map, but are not restricted to a single final Wave.
- **Champions:**
 - are *legendary Bosses* with distinct move sets, potentially phase-based encounters,
 - are intended as **endgame gates**, not standard MVP content,



- can be used to gate higher Difficulties or Tier 3 recipes and similar high-value unlocks.
- Any Wave can, in principle, contain special enemies (Boss, Champion, Elite clusters), but MVP Maps will follow a more predictable escalation pattern to keep learning curves manageable.

6.5 Difficulties and Empowerment

Map Difficulties

- Each Map has multiple **Difficulty levels**.
- Higher Difficulties:
 - increase enemy stats (HP, damage),
 - may unlock additional enemy types or tags,
 - increase rewards (loot quantity/quality, XP, research progress).
- Difficulties are unlocked by:
 - completing the previous Difficulty of the same Map,
 - and/or unlocking via Research (World Tree, Heroes Tree, etc.).
- Lower Difficulties remain **relevant for farming**:
 - each Map has partially *map-specific enemy types*,
 - Research goals may require killing specific enemy families/tags that are more common on lower tiers.

Empower Maps

- **Empower Maps** adds optional modifiers on top of base Difficulty:
 - more monsters,
 - increased enemy stats,
 - additional modifiers expressed as tags (e.g. +Projectiles, +Speed, Leech Aura, etc.).
- Empowerment **costs meta-materials/meta-currency** (e.g. Remains, special tokens):
 - this ties risk/reward tuning to the economy,
 - prevents spamming high-reward setups without cost.
- For MVP, Empowerment is considered an *endgame feature*:
 - can be included with lower priority,
 - base Difficulties should function without Empower being required.



6.6 Live Events (Future)

Map Live-Events are conceptualized as a future extension, not part of the MVP baseline.

Two main variants are envisioned:

- **Variant A – Kill-based Quick-Time Choices (Vampire Survivors-like):**
 - Events triggered based on kill thresholds or Wave milestones.
 - Present the player with quick choices (e.g. temporary buffs, extra enemies for extra loot).
- **Variant B – POE-style Interaction Objects:**
 - In-Map interactables such as strongboxes, rituals, abysses.
 - On click, additional enemy waves or mini-encounters spawn with improved loot rewards.

For MVP these remain out of scope or very limited prototypes; core progression and difficulty should work without Live-Events.

6.7 Rewards and Progression

Per-Wave Rewards

- Regular enemies drop:
 - Parts,
 - Remains,
 - currencies (primarily Gold).
- Drops are handled by the Loot System; Maps define when and where drops are triggered.
- After each successful Wave:
 - XP for surviving heroes is granted,
 - the player may pick up all remaining loot or exit back to the Hideout,
 - progress is saved up to the last successful Wave.

Map Completion Rewards

- Completing all Waves of a Map:
 - marks the Map as completed on that Difficulty,
 - unlocks higher Difficulties for that Map (if available),
 - yields an additional completion reward:
 - * meta-currency (e.g. for Research or Empowerment),
 - * progress in Research or Battlepass-like tracks,
 - * chance to unlock new recipes, heroes or Map variants (depending on design).
- Failure on a Map (all heroes dead) forfeits loot and XP of the current Wave, but does not roll back global progression (research, unlocked Maps, etc.).



6.8 MVP Scope for the Map System

For the initial playable version (MVP), the Map System should deliver:

- **Around 10 Maps:**
 - spread across 3 themes,
 - each Map with distinct layout/obstacle configuration,
 - with fixed but varied Wave structures and enemy mixes.
- **Multiple Difficulties** per Map:
 - at least 2–3 tiers of Difficulty for core Maps,
 - fully playable without Empowerment.
- **Basic Boss implementation:**
 - boosted enemies with more stats and skills.
 - Champions and multi-phase bosses can be postponed or introduced as late MVP/endgame pieces.
- **Empower Maps:**
 - optional, low-priority endgame feature for MVP,
 - only a small, curated set of modifiers initially.
- **Live-Events:**
 - considered *Future*, not core MVP.
 - optional to prototype a single simple event type for testing.

7 Crafting System

7.1 Gear Crafting

7.1.0.1 Overview

The Gear Crafting System combines deterministic player-driven modification (Affixes) with random initial base generation (Implicits). Each gear item is defined by its **Gear Type** (Head, Chest, Leg, Boots, Gloves, Amulet) and can contain:

- Up to **N Implicits** (rolled automatically on creation)
- Up to **M Affixes** (player-crafted and upgradable)

Each mechanic type has its own input/output schema, cost structure, and scaling formula. All crafting transactions are atomic (no partial resource loss).

Access to specific crafting mechanics (Infuse, Re-Infuse, Tempering, Over-Tempering) is gated by the Crafting Research Tree (see Section 4). Early game only basic recipes are available;



advanced mechanics unlock gradually.

Ingredients are always Materials (Parts, Remains). Currencies are account-wide resources (e.g. Gold, Crystals) and are not stored in the item inventory.

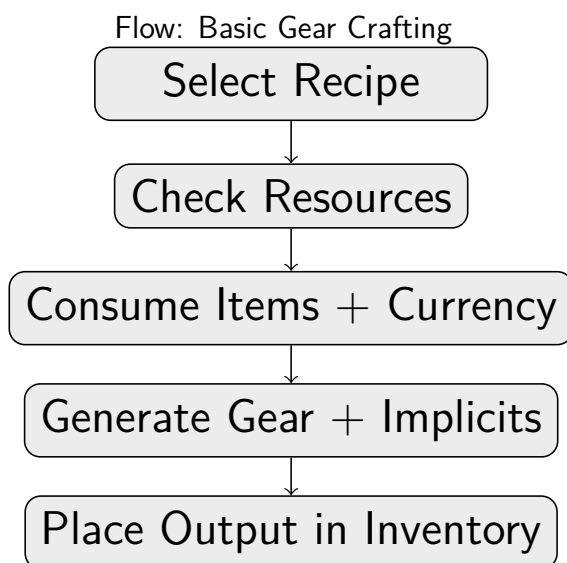
Design Goal: Create meaningful player agency through controlled randomness and resource-driven progression.

7.1.1 Mechanic I – Basic Crafting (Gear Recipes)

7.1.1.1 Purpose

Convert raw materials (and optional boosters) into finished base gear items.

7.1.1.2 Sequence



7.1.1.3 Input / Output Schema

Element	Description / Constraints
Ingredients	Up to 4 Materials (Parts / Remains, each with quantity).
Booster	Max 1 optional Booster Material (typically high-tier Remains).
Currency	Any number of Currencies (Gold, Crystals or similar).
Output	1 Gear item .

7.1.1.4 Cost Structure

- Base cost determined by item rarity and recipe complexity.
- Optional booster adds both cost and potential reward.
- Currency cost acts as sink to regulate player economy.

7.1.1.4.1 Cost Scaling

...

7.1.1.5 Booster Effect

based on Rarity/Quality and Quantity of the used Booster-Material the output is modified.



Parameter	Symbol / Range	Description
Tier Boost per Unit	$f_1(q)$	Adds virtual tier levels to implicits.
Roll Bias per Unit	$f_2(q)$	Shifts random roll toward max value.
Caps	cap_T, cap_B	Hard caps for both effects.
Diminishing Curve	$D(q)$	Function for diminishing returns.

7.1.1.6 Implicits

Implicits and base rolls per Tier

Kurzname	Implicit Effekt	Type	Tag	T1 roll	T2 roll	T3 roll
DMG%	increased damage	%	damage	5–10	10–15	15–25
Thorns	to thorns	flat	damage	1–5	5–7	7–12
Phys+	to phys damage	flat	damage	1–5	5–7	7–12
Elem+	to elemental damage	flat	damage	1–5	5–7	7–12
Armor%	increased armor	%	defence	5–10	10–15	15–25
Res%	increased elemental resistance	%	defence	5–10	10–15	15–25
Dodge%	increased dodge chance	%	defence	5–10	10–15	15–25
Barrier	increased barrier	%	defence	5–10	10–15	15–25
Armor+	to armor	flat	defence	30–60	60–100	100–150
MaxBarrier+	to max barrier	flat	defence	30–60	60–100	100–150
Life%	increased life	%	life	5–10	10–15	15–25
MaxLife+	to max Life	flat	life	20–40	40–75	75–120
Rarity%	increased item rarity	%		5–7	7–10	10–15
Speed%	increased movement speed	%	speed	10–15	15–25	25–40

weight rolls of implicit for each geartype

Stat	Head	Chest	Gloves	Legs	Boots	Amulet
DMG%	100	50	300	100	50	500
Thorns	0	100	50	100	300	0
Phys+	100	50	300	100	50	300
Elem+	300	50	100	100	50	300
Armor%	100	500	50	300	100	100
Res%	300	500	0	300	50	100
Dodge%	300	100	50	500	200	50
Barrier%	300	300	100	100	50	200
Armor+	100	500	50	100	100	50
MaxBarrier+	100	300	100	100	100	50
Life%	100	500	50	100	50	200
MaxLife+	100	500	50	100	100	50
Rarity%	50	25	50	50	100	100
Speed%	0	0	0	0	750	0
sum	1950	3475	1250	2050	2050	2000

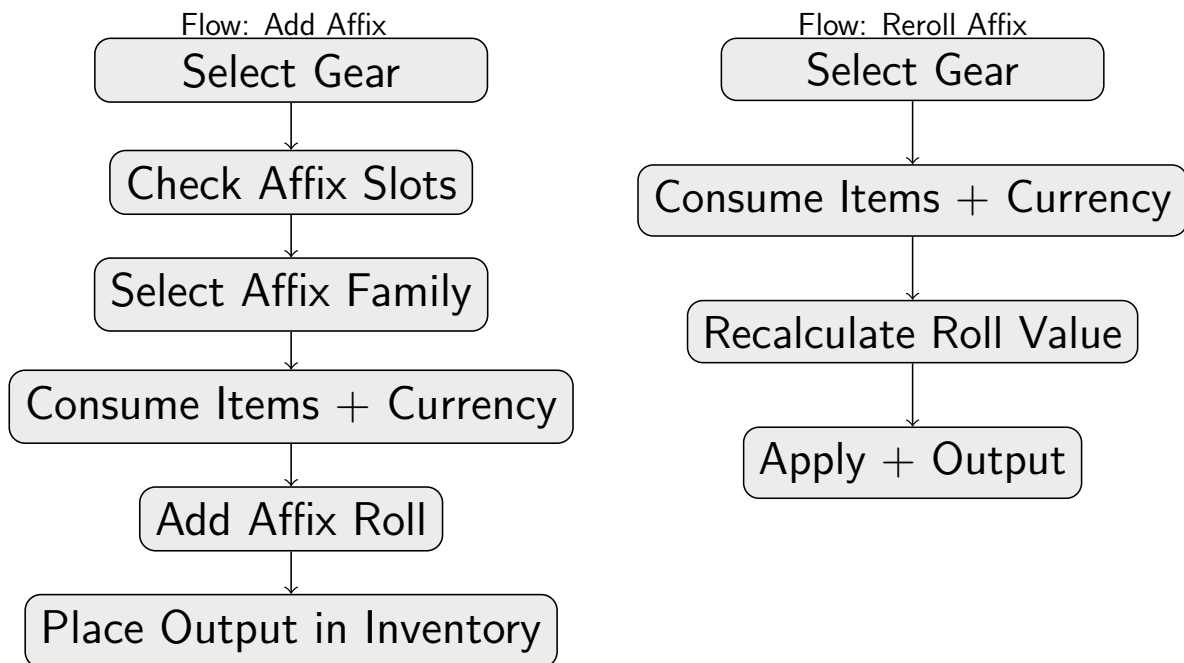
7.1.2 Mechanic II – Upgrade / Affix Crafting

7.1.2.1 Purpose

Allows the player to add new affixes, upgrade existing tiers, or perform one reroll on a designated slot.



7.1.2.2 Sequence



7.1.2.3 Input / Output Schema

Element	Description / Constraints
Base Item	Gear item with available affix slots.
Materials	Crafting reagents depending on action.
Currency	Gold or rare currencies.
Output	Mutated item (same ID, new affix or tier).

7.1.2.4 Cost Structure

- **Add Affix:** base cost plus weighted RNG PickSize (2–4).
- **Reroll:** 1 time per item: high base cost that scales further with the item tier

7.1.2.4.1 Cost Scaling

tbd

7.1.2.5 Affix Pools

7.1.2.5.1 Core Pool

Affix effect	type	tag	T1 roll	T2 roll	T3 roll
to STR	flat	attribute	3–7	7–15	15–20
to DEX	flat	attribute	3–7	7–15	15–20
to CON	flat	attribute	3–7	7–15	15–20
to INT	flat	attribute	3–7	7–15	15–20
to WIL	flat	attribute	3–7	7–15	15–20
increased crit chance	%	crit	10–15	15–25	25–40
increased crit dmg	%	crit	30–60	60–100	100–150
increased phys damage	%	damage	20–40	40–75	75–120
increased elemental damage	%	damage	20–40	40–75	75–120



7.1.2.5.2 Defensive Pool

Affix effect	type	tag	T1 roll	T2 roll	T3 roll
increased all resistance (phys/elem)	%	defence	10–15	15–25	25–40
reduced DoT duration	%	defence	10–15	15–25	25–40
gain % reflect equal to dodge rating	%	defence	20–40	40–75	75–120
to barrier regen per second	flat	defence	3–7	7–15	15–20
reduce barrier recharge delay	%	defence	10–15	15–25	25–40
% dmg taken recouped as life	%	life	1–4	4–8	8–12
leech damage as life	%	life	1–4	4–8	8–12
to life regen per second	flat	life	5–10	10–25	25–40

7.1.2.5.3 Synergy Pool

Affix effect	type	tag	T1 roll	T2 roll	T3 roll
% armor applies to all Dmg	%	damage	10–15	15–25	25–40
% crit chance applies to phys Dmg	%	damage	10–15	15–25	25–40
% dodge rating applies to thorns	%	damage	10–15	15–25	25–40
% Barrier applies to elem damage	%	damage	10–15	15–25	25–40
% phys damage penetrates armor	%	damage	10–15	15–25	25–40
% elem damage penetrates resistance	%	damage	10–15	15–25	25–40

7.1.2.5.4 Utility Pool

Affix effect	type	tag	T1 roll	T2 roll	T3 roll
increased melee skill power	%	skill	20–40	40–75	75–120
increased projectile speed	%	skill	20–40	40–75	75–120
increased area of effect	%	skill	20–40	40–75	75–120
increased ranged skill power	%	skill	20–40	40–75	75–120
increased range	%	skill	20–40	40–75	75–120
increased pierce chance	%	skill	20–40	40–75	75–120
increased cast speed	%	speed	20–40	40–75	75–120
reduced cooldown	%	speed	10–15	15–25	25–40
increased item rarity	%		10–15	15–25	25–40

7.1.2.6 Reroll Affix

- Only one slot per item can be rerolled
- slot is chosen randomly
- Rerolls the Affix effect
- no duplicate of existent Affixes
- the more Affixes are on the item, the smaller the Reroll-Range of which Affix will be rolled
- relativley high base cost, but no Sclaing, since 1 time per item

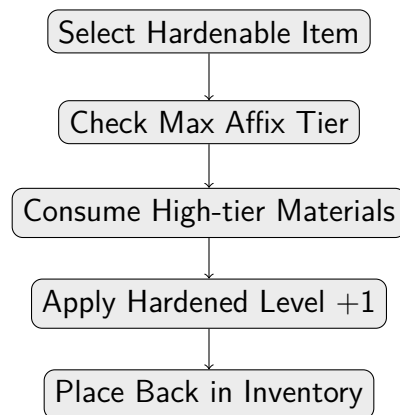
7.1.3 Mechanic III – Hardening (Tier Enhancement)

7.1.3.1 Purpose

A late-game mechanic that permanently increases item potential through tier reinforcement and needs a late-game research to be finished.



7.1.3.2 Sequence



7.1.3.3 Input / Output Schema

Element	Description / Constraints
Base Item	Crafted item with maxed affixes.
Materials	Rare tier-enhancing reagents.
Currency	High-end currency cost (flat + exponential).
Output	Same item with upgraded "Hardened" property.

7.1.3.4 Hardening Tiers

Level	Cost Multiplier	Effect	Notes
1	x1	increase roll within the tier	extreme high cost for late endgame
2	x2	increase roll within the tier	
3	x4	+1 Tier	
4	x5	increase roll within the tier	
5	x6	increase roll within the tier	
6	x12	+1 Tier	
7	x13	increase roll within the tier	
8	x14	increase roll within the tier	
9	x28	+1 Tier	
10	x29	increase roll within the tier	
11	x30	increase roll within the tier	
12	x60	Overroll (over Max)	

7.1.3.5 Cost Structure

- Each hardening level multiplies base material and currency cost.
- Rare materials required (e.g., Essence, Soul Alloy).

7.1.3.5.1 Cost Scaling

see Cost Multiplier in Hardening Tiers

7.1.4 General Validation Rules

- Recipes must define ≤ 4 ingredients and ≤ 1 booster.
- All costs non-negative; zero costs invalid.



- ItemBaseDef must specify GearType and AllowedPools.
- Atomic transactions: no partial resource consumption.
- Rollback mandatory on any failure path.

8 Skill System

8.1 Overview

The Skill System defines how Heroes and Monsters act in combat. It is built on a unified model for Skills, Impacts, Status effects and Modifiers, shared by both sides:

- **SkillData** – base definition: cooldown, range, tags, base damage, target rules.
- **Impacts** – actions executed on cast / hit: deal damage, apply status, trigger other skills.
- **Status** – buffs, debuffs, damage-over-time and similar timed effects attached to units.
- **ModifierData** – generic numeric changes applied by gear, passives, buffs, debuffs, runes, crowns, etc.

Heroes and Monsters use the same SkillData and Status/Modifier framework. The difference lies in where these Skills come from (Hero archetypes, Modules, enemy definition) and which Modifiers apply.

8.2 Shared Stat Model

The Skill System relies on a fixed set of five core stats:

- **STR** – physical power (melee damage, strength-based scaling).
- **DEX** – agility (attack speed, accuracy, evasion-style interactions).
- **CON** – resilience (life, barrier, physical toughness).
- **INT** – mental power (spell power, elemental scaling, complex skills).
- **WIL** – willpower (resistance to control, resource sustain, thematic „will“ effects).

All Skill and Modifier calculations ultimately work on this stat set and derived combat stats (damage, crit, armor, resistances, etc.).

8.3 Skill Model (Heroes and Enemies)

SkillData

Each Skill is defined by a SkillData object, including:

- base damage (physical/elemental or mixed),
- cooldown and cast time,



- range and target rules (single target, AoE, line, cone, self),
- tags (e.g. melee, projectile, DoT, aura, summon, fire, holy, undead),
- scaling hooks (which stats modify what parts of the skill).

Impacts and Status

- **Impacts** define what happens when a Skill is cast or hits:
 - damage instances,
 - application of Status effects,
 - triggering of chained skills (e.g. nova on hit, projectiles on kill).
- **Status** objects represent ongoing effects:
 - buffs (e.g. increased armor, haste),
 - debuffs (e.g. slow, vulnerability),
 - DoTs (e.g. bleeding, burning),
 - other persistent states (e.g. taunt, mark).
- Status effects apply **ModifierData** to stats or skills while active.

ModifierData

ModifierData is the unified language for numeric changes:

- flat additions (e.g. +10 armor),
- additive % increases (e.g. +20% damage),
- multiplicative „more“ multipliers (e.g. 30% more damage),
- caps and clamps (e.g. resistances up to 75%),
- occasional replacements or transformations (e.g. convert physical to fire, replace basic attack).

All systems (Gear, Skill Orbits, Runes, Crowns, Research) ultimately output ModifierData.

8.4 Hero Skill Layers

Hero power and behaviour in combat is defined through several stacked layers:

1. Hero archetype and base stats (STR, DEX, CON, INT, WIL).
2. Gear (from the Crafting System).
3. Skill Modules in Sockets.
4. Socket Orbits (passive progression around each socket).
5. Hero-wide Runes.
6. Endgame Crowns.

Details about Hero archetypes, roles and progression (HeroDef, HeroInstance, AI profiles) are described in Section 11. This section focuses on how Heroes interface with the Skill System via sockets, modules and orbits.



8.4.1 Skill Modules and Sockets

Heroes do not „learn“ traditional skill trees; instead, they use **Skill Modules** socketed into fixed positions.

- Each Hero has **4–5 chakra-style sockets**, not bound to gear:
 - e.g. Torso, Left Hand, Right Hand, Head (+ optional Will/Chest).
- Sockets are **semantically distinct**, not statistically tied to gear slots:
 - e.g. a socket may be associated with crit/burst, sustain, mobility, defence.
- Each socket can hold **exactly one Skill Module item** from the Inventory.
- Socket progression:
 - start with 1 active socket at low level,
 - additional sockets unlock as the Hero levels up (e.g. at level milestones).
- When a Module is socketed:
 - the Hero gains a SkillData instance, based on:
 - * the abstract Module definition,
 - * the Hero archetype (which may skin or specialise the skill),
 - * socket-specific modifiers (see Orbits).

8.4.1.1 Module Role

Modules represent the main active skill layer for Heroes:

- they define **what** a skill does (projectile, aura, buff, summon, etc.),
- archetypes define **how** it feels (e.g. same Module might be a burning nova for an Arcanist and a physical shockwave for a Vanguard),
- Orbits, Gear and Runes further modify behaviour and scaling.

8.4.2 Socket Orbits (Passive Progression)

Each socket has its own **Orbit**: a layered progression ring around the socket that works similar to electron shells in an atom model:

- the *socket* is the core,
- each *orbit tier* is an electron shell around that core,
- orbit *points* are the electrons on that shell.
- an orbit *Mod* is a "Super-electron" (not scientific related) unlocked when electron shell full



8.4.2.1 Structure

- Each socket orbit contains a fixed number of points (e.g. 22 points) spread across multiple tiers (Tier 1, Tier 2, Tier 3).
- Points are **small, incremental stat buffs**:
 - e.g. +% damage, +% life, +crit chance, +status effect, +resource regen,
 - tuned per socket identity (crit/burst vs. sustain vs. utility, etc.).
- Points are invested in free order along the orbit path for that socket.

8.4.2.2 Tier Gates and Orbit Mods

Orbit progression is divided into tiers, each with a gate:

- After investing a required number of points on the current tier (e.g. 6–8 points on Tier 1), the player unlocks a **Tier Mod Choice** for that tier.
- A Tier Mod is a **behaviour/synergy modifier**, not a small stat buff. Examples:
 - convert a single-target skill into a small AoE,
 - add a nova or chain effect on hit,
 - add a DoT component or a mark on hit,
 - convert damage type (e.g. phys to fire) or add penetration,
 - change targeting logic (e.g. prioritise bosses / clusters).
- When a Tier Mod is chosen:
 - its effect is applied permanently to the socketed skill,
 - the next outer orbit tier is unlocked and becomes available for point investment.

8.4.2.3 Socket Identity

Each socket has its own themed orbit and Tier Mod pools:

- **Example (conceptual only):**
 - Torso socket: sustain/defence (life, barrier, mitigation, „on damage taken“ effects).
 - Right Hand: crit and burst (crit chance/multi, „execute“ style effects, high-risk/high-reward).
 - Left Hand: utility/control (crowd control, debuffs, resource manipulation).
 - Head: AoE/range/complex mechanics (larger areas, extra projectiles, complex triggers).
- Tier Mods are drawn from socket-specific pools to reinforce the identity of that slot.



8.4.2.4 Progression Source

Orbit points are gained through level-ups of the Heroe:

- Hero level ups,
- Research unlocks (Skill Tiers in the Heroes Tree),
- possibly specific achievements or milestones (future).

The intended pattern is:

- frequent **small numeric gains** through orbit points,
- occasional **big behavioural steps** when a Tier Mod is unlocked,
- each unlocked Tier Mod both upgrades the current skill and opens access to the next orbit tier.

8.4.3 Crowns (Endgame)

Crowns are an **endgame progression layer**:

- Crown sockets are unlocked via the Heroes Research Tree (see Research System).
- Crowns apply **large, rare, endgame-level modifiers**, such as:
 - transforming a core skill in a fundamental way,
 - adding strong, unique effects with clear downsides,
 - pushing certain stats beyond normal caps.
- Crowns are intentionally rare and limited; they are not part of the early game.

For MVP, Crowns can be treated as *future or late* content and do not need full implementation.

8.5 Skill Loadout and Combat Usage

Hero Loadout

A Hero's combat loadout is defined by:

- the number of **active sockets** (unlocked by level),
- which **Modules** are slotted into those sockets,
- which **Orbit mods** have been chosen around each socket,
- hero-wide **Runes** and possible **Crowns**,
- equipped Gear and its interactions with skill tags.

In practice:

- early game: 1 active socket → 1 major skill per Hero,
- mid game: 2–3 sockets → 2–3 major skills per Hero,
- late game: 3–4 (5) sockets with fully developed Orbits.



AI-Driven Skill Usage

There is no direct player control over skill priority in combat:

- each Hero follows an **archetype-specific AI profile**:
 - tanks prioritise taunts/defensive skills and high-HP targets,
 - damage dealers prioritise high-value or low-HP targets,
 - support archetypes prioritise buffs and heals.
- AI rules consider:
 - skill tags (e.g. „heal“, „buff“, „AoE nuke“),
 - cooldown states,
 - simple conditions (e.g. ally HP, enemy clustering).

The player expresses build intent primarily through:

- module choices (which skills exist at all),
- orbit mods (how those skills behave),
- gear and runes (which stats and tags are favoured).

8.6 Interactions with Other Systems

Research System

- The **Heroes Research Tree** unlocks:
 - new Hero archetypes (Hero packs),
 - deeper Skill tiers (access to higher Orbit tiers),
 - Crown sockets as endgame goals.
- Research does not grant direct one-run buffs, but expands long-term build options.

Gear and Crafting

- Gear affects Skills primarily via:
 - stats that scale skills (STR, INT, crit, penetration),
 - tag-based bonuses (e.g. increased projectile skill power, increased AoE, reduced cooldown).
- The Crafting System defines how gear is created and upgraded; Skill System uses the resulting stats and tags.

Inventory and Modules

- Skill Modules and Runes are inventory items:
 - Modules and Runes are managed in the Hideout inventory screen,
 - equipping is done via sockets on the Hero panel.
- No inventory management is available during Maps; skill loadouts are fixed for the duration of a run.



Map & Combat System

- The Map & Combat System calls into the Skill System to:
 - execute Skills for Heroes and Monsters,
 - resolve Impacts and Status effects,
 - track kills, damage and other metrics for Research.

8.7 MVP Scope for the Skill System

For the MVP, the Skill System should deliver:

- A stable **SkillData / Status / Modifier** framework used by Heroes and Monsters.
- A limited set of **Hero archetypes** (e.g. 3–5) with:
 - clear roles,
 - 1–3 key Skill Modules each,
 - basic Orbit layouts (Tier I/II, Tier III optional).
- **Sockets and Modules:**
 - start with 1 active socket per Hero,
 - unlock up to 2–3 sockets for MVP,
 - small, strongly themed Module pool.
- **Orbits:**
 - functional Tier I/II for each socket,
 - Tier III and complex mutations may be endgame or future additions.
- **Runes:**
 - optional, simple hero-wide bonuses (1 slot is sufficient for MVP),
 - complex interactions can be postponed.
- **Crowns:**
 - treated as late-game or future content,
 - only minimal scaffolding needed in MVP.

9 Loot and Drop System

9.1 Loot Flow and Rarities

9.1.0.1 Overview

This subsection defines the generic loot flow from enemy death (and wave resolution) to final rewards. It connects the Map & Combat System with the Inventory System and the Item/-Material categories (Items: Gear, Modules, Runes; Materials: Parts, Remains; Currencies: Gold, Crystals), but is implementation-agnostic and focuses on inputs, order of operations and balancing knobs.



9.2 Loot and Rarities

Purpose

This section specifies the complete loot flow from wave setup to final rewards. It is implementation-agnostic and defines inputs, order of operations, and balancing knobs.

Key Terms

- **Wave Budget** B : Total loot value budget allocated to a wave.
- **Spent Budget** U : Accumulated loot value already awarded in the current wave.
- **Rarity** R : Item quality tier (e.g., Common, Uncommon, Rare, Epic, Legendary, ...).
- **Base Chance** $p_{\text{base}}[R]$: The configured per-rarity drop chance before any modifiers.
- **Unlucky Multiplier** $m_{\text{unlucky}}[R]$: Pity-protection factor derived from recent fails / hits per rarity.
- **Luck Factor** L : Linear luck strength; $L = 0$ (no bonus), $L = 1$ (+100).
- **Luck Scale** $k[R]$: Optional per-rarity scaling of luck (default $k[R] = 1.0$).
- **Budget Multiplier** $m_{\text{budget}}(U, \text{lootValue}, B, R)$: Soft damping when U approaches/exceeds B , including rarity floors.
- **Effective Tags**: The union of an enemy's *baseTags* and *bonusTags*.

0) Wave Setup

1. **Compute Wave Budget** B : Derived from wave composition (ranks, counts), map difficulty and level; add variance; initialize $U = 0$.
2. **Load and Merge Loot Rules**: For all tags present in the wave, prepare the merged rule set (drop pools, min/max drops, rarity guarantees, secret rules).

1) On Enemy Kill

1. **Determine Roll Count**: Based on enemy rank/type (e.g., Elite may trigger more rolls than Normal).
2. **Build Effective Tag Set**: $\text{effectiveTags} = \text{baseTags} \cup \text{bonusTags}$ (duplicates ignored for logic; weighting optional).

2) Per Roll (for this enemy)

1. **Pick Tag**: Randomly select one tag from *effectiveTags* (optionally bias *bonusTags* slightly).
2. **Resolve Tag Rule**: Activate the corresponding merged loot rule for this tag (yields a list of loot entries to attempt).

3) Iterate the Tag's Drop Entries

For each *loot entry* (which may be a single chance or a list of chances), evaluate one *rarity block* at a time. The system supports *parallel rarity rolls* so that multiple rarities can succeed in the same slot.



Factor Order (per rarity R):

1. **Base Chance:** $p_{\text{base}}[R]$
2. **Unlucky (Pity):** multiply by $m_{\text{unlucky}}[R]$
3. **Luck (Linear, No Cap):** multiply by $(1 + L \cdot k[R])$ *Example:* $p_{\text{eff, preBudget}}[R] = p_{\text{base}}[R] \cdot m_{\text{unlucky}}[R] \cdot (1 + L \cdot k[R])$
4. **Budget (Soft Brake + Floors):** multiply by $m_{\text{budget}}(U, \text{lootValue}, B, R) \rightarrow p_{\text{roll}}[R] = p_{\text{eff, preBudget}}[R] \cdot m_{\text{budget}}(\cdot)$
5. **RNG Roll:** If success, **award one item** of rarity R (this slot can yield multiple items if several rarities succeed).

Post-Roll Accounting (per rarity):

- On success: add the item to results; increase U by its `lootValue`; call `Unlucky.OnDrop( $R$ )`.
- On fail: call `Unlucky.OnFail( $R$ )`.

4) Secret Drops (Optional, Per Enemy)

Evaluate secret/conditional drops (e.g., specific tag combinations). These are additional rolls that can yield extra items. They do not interfere with the regular pity counters unless explicitly specified.

5) Wave Finalization

1. **MinDrops Failsafe:** If total items $<$ configured minimum for the wave, add items from the wave's mixed pool until the minimum is met. This path increases U ; whether it resets pity per rarity is a policy toggle (recommended: *does not* reset pity to preserve fairness).
2. **Rarity Guarantees:** If a minimum per rarity is required, top up those rarities. This path typically *does not* alter pity and may optionally bypass budget damping depending on economy policy.

Design Notes

- **Parallel Rarity Rolls:** Each rarity is evaluated independently; *all successes drop*. Lower-tier success does not consume or block higher-tier success.
- **Luck Scope:** Luck affects *quality* (rarity chances) only. Quantity remains governed by entries, rules, and budget. If desired, allow certain luck-driven drops to *ignore* budget damping (economy-sensitive).
- **Budget as Soft Brake:** m_{budget} smoothly reduces chances when U exceeds B ; rarity floors prevent near-zero results for higher tiers.
- **Tags as Theme Driver:** Use *effectiveTags* for both regular and secret rolls; optional small biases can keep `bonusTags` slightly more impactful without excluding `baseTags`.



Balancing Knobs

- **Per-rarity base chances** $p_{\text{base}}[R]$
- **Unlucky growth/reset rates** per rarity
- **Luck factor** L (from passive tree, map mods, etc.) and **luck scales** $k[R]$
- **Budget damping** curve parameters and **rarity floors**
- **Tag pick weighting** (base vs. bonus) and secret-drop criteria
- **Failsafe policies**: whether MinDrops/Guarantees affect pity or budget

10 Meta Progression and Battlepass

10.1 Overview

The Meta Progression / Battlepass System provides long-term goals across multiple sessions. It runs in parallel to the core progression systems (Research, Heroes, Crafting, Maps) and rewards regular play with additional resources.

This system is **not monetised** (no paid track). It is designed purely as a long-term motivation layer and a light „quest wrapper“ around the existing gameplay.

10.2 Design Goals

- **Structured Long-Term Goals**: Provide clear, mid-term targets beyond „more damage / higher difficulty“.
- **Soft Onboarding / Tutorial Replacement**: Early Battlepass objectives can guide new players through key systems (Maps, Research, Crafting, Heroes) instead of a rigid tutorial.
- **Daily / Weekly Engagement**: Offer small, optional tasks on a daily/weekly basis for players who enjoy checklists and routines.
- **Non-Essential Power**: Rewards support progression (materials, currencies) but must not be the only source for critical power.

10.3 Structure

The Meta Progression is structured as a **season-neutral** track with recurring objectives:

- **Core Track (Account-Wide)**:
 - A linear progression bar with levels.
 - Progress gained by completing Maps, Waves and simple objectives.
 - Each level grants a small reward bundle.
- **Objectives / Quests**:
 - Simple, gameplay-aligned goals, such as:



- * „Defeat X enemies on Map type Y.“
- * „Complete N Waves without losing a Hero.“
- * „Craft or upgrade an item.“
- Objectives exist in three scopes:
 - * **Intro Objectives:** early, tutorial-like tasks introducing core systems.
 - * **Daily Objectives:** small tasks meant to be completed in a short session.
 - * **Weekly Objectives:** slightly larger goals for regular players.
- Completing objectives grants Battlepass XP (progress on the core track) and sometimes direct rewards.

10.4 Rewards

Battlepass rewards should be „nice to have“, not mandatory:

- **Primary Rewards (Economy-Oriented):**
 - rare Materials (higher-tier Remains, special crafting reagents),
 - rare Currencies,
 - Gold bundles.
- **Secondary Rewards (Future / Optional):**
 - cosmetic variations (skins, visual effects),
 - cosmetic Hideout decorations or UI themes.

The Battlepass must not be the only reliable source for:

- key unlocks (Maps, Heroes, basic crafting),
- core power spikes (Gear tiers, base stat growth).

It is an *accelerator* and a „reward wrapper“ around what the player already does, not a mandatory grind path.

10.5 Interaction with Other Systems

- **Research System:**
 - Objectives can reference Research milestones (e.g. „Start a new research project“, „Complete X kills for research“),
 - but Research unlocks remain independent from Battlepass progression.
- **Map System:**
 - Many objectives are Map/Wave based (kills, completions, difficulty tiers).
 - Map variety (themes, enemy families) can be reflected in specific objectives.
- **Crafting and Inventory:**
 - Objectives can reward and require basic crafting actions (craft an item, infuse gear),
 - Reward bundles contain Materials and currencies that directly feed into crafting.



10.6 MVP Scope

For MVP, the Battlepass / Meta Progression should remain simple:

- **Single Core Track:**
 - one account-wide bar with a limited number of levels,
 - no seasonal reset logic required initially.
- **Basic Objective Sets:**
 - a small set of Intro Objectives that guide the player through:
 - * playing Maps,
 - * using the MapDevice,
 - * starting Research,
 - * crafting basic gear,
 - * equipping Heroes.
 - a limited pool of Daily/Weekly objectives rotating randomly.
- **Reward Types:**
 - bundles of Gold,
 - mid-tier Materials,
 - small amounts of rare Currencies.

Further extensions (cosmetics, advanced objective chains, seasonal structure) can be added once core progression and economy are stable.

11 Heroes and Archetypes

11.1 Overview

Heroes are the core units the player builds, equips and sends into Maps. Each Hero belongs to an **Archetype** which defines their role, core stats, AI behaviour and passive traits. Heroes do not have traditional skill trees; instead, they use the Skill System via sockets, modules and orbits (see Section 8).

- **Archetype** = role, stat focus, base behaviour, passive identity.
- **Hero** = instance of an archetype with a level, gear, skill modules, orbits, runes and crowns.
- **Team** = up to 4 Heroes fielded together in combat.

11.2 Team Size and Hero Unlocks

Team Size

- The player starts with a **single Hero slot** (team size 1).
- Additional team slots are unlocked over time (via progression or research) up to a maximum of **4 Heroes per team**.
- The game is balanced around 1–4 Heroes auto-battling together.



Hero Unlocks

Heroes are not all available from the start:

- The player starts with exactly **one archetype** (starter Hero).
- Additional Heroes are unlocked in **packs of 2–3 archetypes**, themed together (e.g. „ranged pack“, „support pack“).
- Unlocking and owning are separate:
 - *Unlock* makes an archetype available to be recruited.
 - *Recruitment* may cost a rare currency.
- The first few (approx. 5) Heroes are granted through story/progression without heavy cost.
- Further Heroes are primarily unlocked via the Heroes Research Tree and then purchased with rare currency.

11.3 Archetype Roster

Table 8 lists the planned archetypes, their primary role and core stat focus. This roster is larger than the MVP set; the MVP subset is marked explicitly.

For MVP, the core playable set is: **Vanguard, Arcanist, Ranger, Summoner**. Additional archetypes are planned for later iterations once core systems are stable.

11.4 Stats and Growth

All archetypes share the same five core stats:

- STR, DEX, CON, INT, WIL (see Section 8).

Each archetype defines:

- 1 primary **Core stat** (e.g. CON for Vanguard, INT for Arcanist),
- 1–2 secondary stats,
- remaining stats as low-impact for that archetype.

Exact numeric growth per level and per-stat distributions are defined in the balance sheets. For the GDD, it is sufficient that:

- Vanguard and Bruiser lean heavily into **CON/STR**,
- Ranger and Marksman lean into **DEX** (with some INT or STR depending on design),
- Arcanist, Alchemist, Saboteur rely on **INT**,
- Summoner, Hierophant, Oracle rely on **WIL**.



11.5 Archetype Passives

Each archetype has a small set of defining passive traits. These are always active once the Hero is in the team and help differentiate playstyle and build direction.

Vanguard (Tank, CON Core)

- **Bulwark:** +20% Armor.
- **Stand Your Ground:** reduced impact from slows / stuns and/or can move through enemies without being body-blocked (exact implementation to be decided).
- **First Hit Guard:** the first hit taken from each enemy is reduced by 50% (subject to implementation feasibility).

Piercer (STR Core, Execute)

- **Deadly Precision:** +50% critical damage multiplier.
- **Hemorrhage:** unlocks a stackable Bleed DoT effect on hits.
- **Merciless:** enemies below 40% HP take +15% damage from Piercer.

Arcanist (INT Core, DoT/Control)

- **Arcane Echo:** +25% debuff duration.
- **Overchannel:** +20% spell damage.
- **Flow of Mana:** skills have -15% cooldown.

Ranger (DEX Core, Projectiles/Kiting)

- **Volley:** +1 projectile count for applicable skills.
- **Through the Line:** +25% pierce chance (projectiles can hit multiple enemies).
- **Hunter's Focus:** +15% range.

Marksman (DEX Core, Precision)

- **Headshot:** +20% critical chance.
- **Longshot:** +20% damage against targets further than a threshold distance (e.g. 8m).
- **Sniper Discipline:** +10% attack speed while stationary (after not moving for a short duration, e.g. 5s).

Alchemist (INT Core, DoTs/Zones)

- **Virulent Concoction:** DoTs can stack +10 additional times.
- **Toxic Spread:** on kill, existing DoTs can spread to a nearby target.
- **Lingering Poison:** +25% DoT duration.



Summoner (WIL Core, Minions)

- **Broodmaster:** +2 summon count (base summon count = 3).
- **Empowered Minions:** minions have +20% HP and +15% damage.
- **Hive Mind:** +5% damage per living minion.

Hierophant (WIL Core, Support/Nature)

- **Blessing Aura:** allies in a small radius (e.g. 4m) regenerate 1% HP/sec.
- **Sanctuary:** the next attack against nearby allies is negated (cooldown, e.g. every 10s).
- **Spirit Bond:** buffs applied by Hierophant last +25% longer.

Bruiser (STR Core, Sustain Melee)

- **Brawler's Might:** +20% attack damage.
- **Blood for Strength:** +10% life leech factor.
- **Thick Skin:** +15% maximum HP.

Saboteur (INT Core, Traps/Procs)

- **Tinker's Reserve:** traps have +1 charge.
- **Suspicious Gear:** increased chance to apply additional DoT stacks.
- **Expertise:** increased area of effect for trap-triggered effects.

Oracle (WIL Core, Buff/Debuff)

- **Prophetic Weakness:** enemy debuffs last +25% longer.
- **Battle Clairvoyance:** allies in a radius (e.g. 6m) gain +10% critical chance.
- **Foretold Resilience:** allies in a radius gain +10% elemental resistance.

11.6 Sockets and Orbit Progression (Hero Perspective)

Details of sockets, modules and orbits are defined in the Skill System (Section 8). From the Heroes perspective, the important points are:

- Each Hero has a fixed set of **4–5 skill sockets** (chakra-style, not tied to gear).
- The Hero starts with **1 active socket** at level 1.
- Additional sockets unlock as the Hero levels up:
 - all sockets should be unlocked by approximately **level 60**,
 - the remaining levels (60–100) primarily increase orbit progression and prepare for the Crown system.
- Each level grants orbit points that can be spent on socket orbits; exact points-per-level and bonus sources are defined in the Skill/Balance sections, but the intent is:



- base: around 1–2 orbit points per level,
- additional points from Map milestones and socket set completion,
- a „pure“ level 100 build should be able to complete the intended orbit path.

Level 100 marks the end of regular Hero progression: after that, only Crowns and meta systems extend power.

11.7 AI Profiles (Behaviour)

Heroes are fully AI-driven in combat. The player does not manually set skill priorities. Instead, each archetype has a characteristic AI profile:

- **Vanguard / Bruiser:**
 - prioritise enemies that threaten allies or are closest to the front,
 - use defensive and control skills early in fights,
 - stay near the main enemy cluster.
- **Piercer / Marksman / Ranger:**
 - prioritise low-HP or high-value targets,
 - maintain distance when possible (Ranger/Marksman),
 - use burst skills on high-HP targets or bosses.
- **Arcanist / Alchemist / Saboteur:**
 - prioritise clustering (AoE) and debuff application,
 - use control skills on elites/bosses when available,
 - Saboteur places traps ahead of enemy movement paths.
- **Summoner:**
 - maintains minion count as a priority,
 - uses skills that buff or reposition minions when possible,
 - stays behind frontline Heroes.
- **Hierophant / Oracle:**
 - prioritise healing and shielding low-HP allies,
 - apply buffs early in encounters and keep them up,
 - apply key debuffs to high-HP or elite targets.

Exact AI rules, thresholds and target selection weights are part of the Technical Design Document, but this section defines the intended behaviour per archetype from a design perspective.



11.8 MVP Hero Set

For the MVP, the target is:

- **4 core archetypes** fully implemented:
 - Vanguard (tank),
 - Arcanist (full caster),
 - Ranger (bow/ranged fighter),
 - Summoner (minion-focused).
- Each with:
 - a functional set of 2–3 key Skill Modules,
 - full socket unlocks by around level 60,
 - a meaningful orbit layout (Tier I/II, with Tier III and Crowns optional as late/endgame).
- Additional archetypes (Marksman, Alchemist, Hierophant, Bruiser, Saboteur, Oracle) are planned as **post-MVP extensions**.

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System	Inputs	Outputs	Core Responsibility
Research System	XP (global), Kill-Counts (type/tag-sensitive), Map completions	Unlocks (Hideout elements, Heroes, Item tiers, Crafting options, Map difficulties)	Long-term meta-progression and gated introduction of new systems and content.
Skill System	Hero XP, build decisions (skill points, passives)	Active skills, passive effects, modified skill behaviour	Defines what each Hero can do in combat (actives) and how they scale/modify via passives.
Crown System	Endgame resources, late-game Hero progression	Endgame skill boosts / enhancements	Provides an additional endgame layer of powerful, limited boosts on top of the base Skill/Passive System.
Inventory System	Loot drops (from Maps), rewards (Research / Battlepass), crafting outputs	Items (gear, materials), currencies (gold, rare currencies)	Persistent storage and organisation of all tangible player resources (items & currencies).
Crafting System	Recipes, base items, currencies/materials	Upgraded / transformed items	Transforms input items and currencies into stronger or altered items via Recipes, Upgrading and Hardening.
Map & Combat System	Selected Heroes (team), their skills, gear, current Map settings	Combat results: Loot, XP per Hero, Map completion state	Runs auto-battle encounters (Waves) on a selected Map and determines victory/defeat, drops and XP.
Loot System	Combat events (enemy deaths, chests, Map rewards)	Generated loot items, gold, rare currencies	Decides what drops when and with which rarity/quality; feeds the Inventory and Crafting Systems.
Meta Progression / Battlepass	Playtime, completed Maps/Waves, milestones (objectives)	Cosmetic rewards, account-wide bonuses	Parallel long-term track that rewards regular play with cosmetics and light meta rewards.

Table 4: Core Systems Overview



Node	Short Description	Unlocks / Effects
Basic Gear	<i>„Let’s see what we can do with these shreds.“</i>	Unlocks basic gear usage and simple item crafting from common loot. Establishes the baseline of the item system.
Infuse I	<i>„We can make use of these magical remains.“</i>	Unlocks the first tier of infusion: adding simple magical properties to base items using low-tier materials.
Re-Infuse	Allows re-rolling or replacing existing infusions at a cost.	Unlocks re-infusing items, enabling early deterministic rerolls instead of pure item hunting.
Gear II	<i>„Find some of the rarer materials, so we can improve the gear.“</i>	Unlocks access to higher gear tiers and rarer crafting materials; raises the ceiling for item power.
Refine / Tempering	Introduces finer control over item stats via tempering.	Unlocks refining/tempering operations that adjust existing stats within defined ranges.
Infuse II	Second tier of infusion for more powerful effects or additional affix slots.	Unlocks stronger infusions and/or more slots per item, increasing build complexity.
Gear III	Unlocks late-game gear tiers.	Gives access to high-end gear bases and top-tier materials for endgame items.
Infuse III	Third tier of infusion, oriented towards endgame builds.	Unlocks the highest tier of infusions with strong, build-defining properties.
Over-Tempering	Extreme, high-risk/high-reward tempering.	Unlocks over-tempering mechanics that allow pushing items beyond normal limits at significant cost or risk.

Table 5: Crafting Research Tree (MVP nodes)



Node	Short Description	Unlocks / Effects
Empower Maps	Enables basic Map empowerment.	Unlocks modifiers that increase Map difficulty in exchange for better rewards (e.g. more enemies, higher damage/HP, better loot).
Map Live-Events	Introduces dynamic events during Maps.	Unlocks live events during a Map (e.g. reinforcements, shrines, hazards) that alter Wave flow and risk/reward mid-run.
Undead Monsters	Expands the enemy ecosystem with undead.	Unlocks undead-themed enemy packs, bosses and tags; adds new resistances/weaknesses and synergies.
Holy Monsters	Introduces holy/fanatic enemy types.	Unlocks holy-themed enemies and factions with their own mechanics and counters, further diversifying encounters.

Table 6: World Research Tree (MVP nodes)

Node	Short Description	Unlocks / Effects
Hero Packs (Story / Theme)	Unlocks groups of Heroes along narrative or thematic lines.	Frees 2–3 Heroes at a time, grouped by story or theme, instead of single isolated Heroes. Encourages experimenting with new team archetypes.
Skill Tiers	Unlocks deeper layers of the skill trees.	Opens higher skill tiers (e.g. Tier 2/3 nodes) for existing Heroes, enabling more specialised builds and advanced synergies.
Crown Sockets	Prepares Heroes for the endgame Crown system.	Unlocks Crown sockets on Heroes or their skill trees, allowing equipping powerful endgame Crown boosts once available.

Table 7: Heroes Research Tree (MVP nodes/concepts)



Archetype	Role	Core Stat	Identity (Short)	MVP
Vanguard	Tank / Frontline	CON	High durability, control, damage mitigation, „front wall“.	Yes
Piercer	STR DPS / Execute	STR	Physical single-target burst, armor penetration, executes.	Future
Arcanist	Caster / Control	INT	Elemental damage, DoTs, debuffs and control.	Yes
Ranger	Ranged Hybrid	DEX	Mobile ranged attacker, projectiles, kiting tools.	Yes
Marksman	Ranged Single-Target	DEX	Long-range precision, crit-focused, positional bonuses.	Future
Alchemist	DoT / Zone Control	INT	Damage-over-time, ground effects, debuffs.	Future
Summoner	Minion Master	WIL	Summoned allies, scaling with minion count and survival.	Yes
Hierophant	Support / Nature	WIL	Healing, auras, protection, buff duration.	Future
Bruiser	Sustain Melee	STR	Melee DPS with self-sustain and high HP.	Future
Saboteur	Traps / Procs	INT	Traps, delayed damage, proc-based synergies.	Future
Oracle	Buff / Debuff Support	WIL	Team-wide buffs and enemy debuffs, predictive flavour.	Future

Table 8: Archetype roster and core roles